

Aditya Raj

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EDUCATION

Indian Institute of Information Technology, Tiruchirappalli
BTech in Electronics and Communication Engineering (CGPA: 7.12 / 10)

Aug 2023 - May 2027
Tiruchirappalli, Tamil Nadu

ACHIEVEMENTS

- **LeetCode: Guardian** (Max Rating: **2192**, Top 0.2% Globally) – Ranked **155th** & **331st** in Weekly Contests 438 & 445 among 32,000+ participants (@adityaaarajjj).
- **Codeforces: Expert** (Max Rating: **1845**) – Ranked **316th** & **403rd** in Div. 2 (Rounds 1108 & 1064), and **133rd** in Div. 3 (@adityaraj18).
- **CodeChef 4★** (Max Rating: **1837**, @adityaaa_rajjj) and **6-Kyu at AtCoder** (Rating: **980**, @de_barr_).

TECHNICAL SKILLS

Languages: C++, Python, JavaScript (ES6+), TypeScript, Java, HTML5, CSS3, SQL
Frontend: React.js, Next.js, Redux, Tailwind CSS, Modern CSS, HTML Canvas, Vite
Backend & Databases: Node.js, Express.js, REST APIs, WebSockets, MongoDB, PostgreSQL
Tools & Concepts: Git, GitHub, Linux, Docker, VS Code, Postman, OOP, System Design, DBMS, OS, Computer Networks

PROJECTS

CodeAudit | React, Node.js, Express, Gemini, Docker, Redis, MongoDB GitHub 

Real-Time AI-Powered Code Review & Repository Analysis Platform

- Built a full-stack platform for real-time GitHub repository analysis, combining AI-powered code review with execution-based code health analysis.
- Integrated Google Gemini for contextual code reviews, CWE detection, and repository-aware assistance with real-time WebSocket streaming.
- Engineered an asynchronous BullMQ + Redis pipeline with isolated Docker sandboxes for secure code execution with CPU, memory, timeout, and network restrictions.
- Implemented GitHub REST API integration with Redis ETag caching, MongoDB-backed analysis history, and historical Code Health Score tracking across commits.

CP-Tracker | React, Node.js, Express, MongoDB, JWT, Cheerio GitHub 

Unified Competitive Programming Stats Dashboard & Public Profile Platform

- Built a full-stack dashboard aggregating competitive programming stats across LeetCode, Codeforces, CodeChef, AtCoder, and GeeksforGeeks into a single unified view.
- Designed a zero-OAuth handle verification system using bio-injection and submission-token proofs to confirm platform handle ownership.
- Engineered a cron-based background sync service refreshing verified stats every 6 hours, with anomaly detection to guard against erroneous rating/solve-count drops.
- Implemented JWT-based authentication with bcrypt password hashing and shareable, zero-auth public profile pages exposing only verified stats.

AI Task Manager | Node.js, Express.js, MongoDB, React, REST APIs, NLP GitHub 

Full-Stack Intelligent Autonomous Task & Project Management Application

- Engineered an AI-powered project management platform featuring autonomous goal decomposition into structured milestones and subtasks.
- Implemented an NLP-driven Command Palette (Cmd+K) parsing natural language strings for instant task creation with priority, tags, and due dates.
- Built interactive workflows including HTML5 drag-and-drop Kanban board, 4-quadrant Eisenhower Matrix, and productivity velocity analytics.
- Integrated a Pomodoro focus engine with Web Audio API sound synthesis and created a modular Express REST backend with atomic data backups.

RELEVANT COURSEWORK

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|--------------------------------|---------------------|------------------------------|-----------------------|
| • Data Structures & Algorithms | • Operating Systems | • Database Management (DBMS) | • Computer Networks |
| • Object-Oriented Programming | • Machine Learning | • Artificial Intelligence | • Digital Electronics |